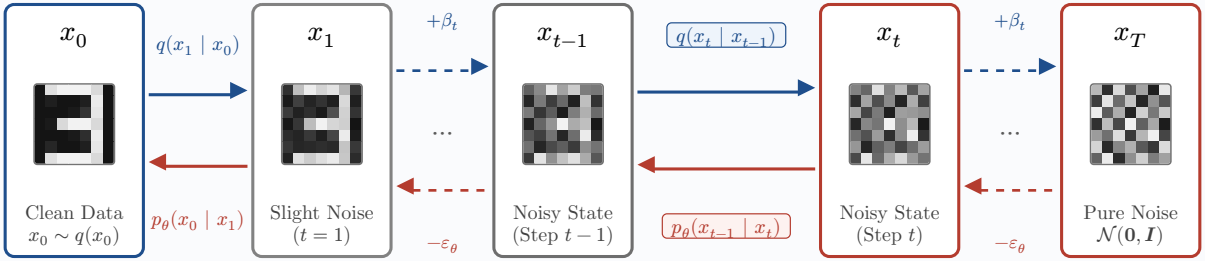
FORWARD NOISING PROCESS (Fixed Markov Chain)

Step transition $(\beta_1, ..., \beta_T)$: $q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}x_{t-1}, \beta_t \boldsymbol{I}\right)$

Reparameterization Form

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$$

Closed-Form Direct Jump (1 Step): $q(x_t \mid x_0) = \mathcal{N}\left(\sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)\boldsymbol{I}\right)$



REVERSE DENOISING PROCESS (Learned Sequential Chain)

Iterative step $(T \to 0$, no skip$)$: $p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 \boldsymbol{I})$

U-Net Noise Predictor $\varepsilon_\theta(x_t, t)$

$$L_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right]$$